



System Subsystem Specification
for the
The Balance Project
DCN: SDP20231224-P1:128
Revision Date: 06 Dec, 2025

Prepared by:

Vinay Agarwal
Balance Project

Controlled by: AGENCY
Controlled by: OFFICE
CUI Category: WILL-BE
Limited Dissemination Control: TBD
POC: Undefined Gpl, undefined.l.gpl.civ@us.navy.mil

Distribution Statement D: Vinay Agarwal



DOCUMENT CHANGE HISTORY

The following table is a simple list of released revisions sent for review. Records of reviews and the review artifacts are saved with reviewer information in the The Balance Project artifact repository.

Change Record

Date	Version	Author(s)	Change Reference
13 Sep 2025	P1	Vinay Agarwal	Preliminary DRAFT version

Draft P1 Preliminary version of this document.

1. Baseline Document draft 1



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CHAPTER 1

Scope

This document provides the System / Subsystem Specification (**SSS**) for the Balance System. The system will be referred to as the Balance System.

Acronym, used for testing, is **.PDF**. Acronym, used for testing, is **BioAPI**. Acronym, used for testing, is **MISO**. Acronym, used for testing, is **UAV**. Acronym, used for testing, is **TOC-L**. Acronym, used for testing, is **USN**. Acronym, used for testing, is **USAF**. Acronym, used for testing, is **USSF**. Acronym, used for testing, is **USMC**. Acronym, used for testing, is **USArmy**.

Acronym, used for testing, is **Garmin**. Acronym, used for testing, is **JIDA**.

1.1 Identification

The Balance System described in this document shall be known as Balance System version 1. However, the **SSS** described herein shall be applicable to pre-releases such as Beta-releases for a phased release as listed for each requirement. The major system interfaces and capabilities are fully specified in Chapter 3.

1.2 System Overview

Figure 1 shows the high-level architecture for the Balance System system. This diagram shows the major external interfaces that provide the capabilities of Balance System.

The Balance System is a single player game. The game is a standalone game played on the STM32 board. This system would be a game where the user would have to balance a ball on a LCD screen that is builtin on the STM32 board. The objective of the game is to balance the ball on the screen based on the way the board was tilted. Using an onboard gyroscope that communicates across the board to the MCU though SPI communication. Balance System would keep track of the current position of the ball and where the next updated move is. This helps keep track of the system of where the ball is until a movement has occurred. Balance System shall process at a maximum 180 Hz. This would give the user enough time to process the current angle of the ball and be able to present on the LCD-TFT screen.

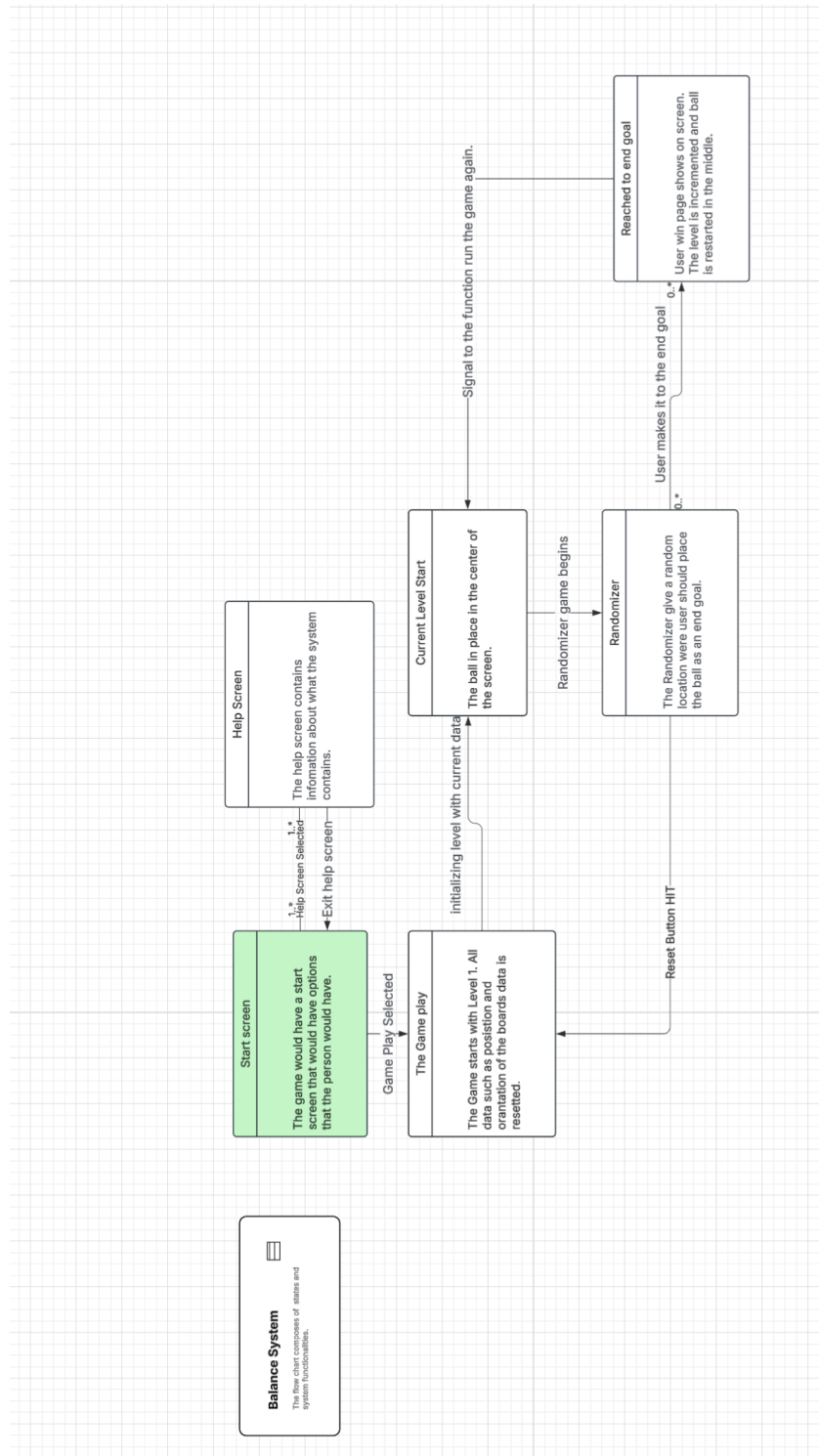


Figure 1: System Overview



1.3 Document Overview

This section provides information about this document's security/privacy considerations, contents, structure, and version information. This section also provides information regarding how specifications are formatted in this artifact and how they can best be understood.

1.3.1 Security and Privacy Considerations

This document has been identified as "Controlled Unclassified Information" (CUI). Please follow the control block on the title page for ownership, creation, category, dissemination, and Point of Contact (**POC**) information. This information should be delineated, per <https://www.dodcui.mil> as:

Owner the name of the DoD Component (not required if identified in the letterhead)

Creator identification of the office creating the document

Category identification of the categories contained in the document

Dissemination applicable distribution statement or limited dissemination control (LDC)

POC name and phone number or email of POC

1.3.2 Artifact Format

This document format is based upon the guidance in the **SRD** template from MIL-HDBK-520A [**ref`MIL`HDBK`520**]. The specifications and associated acceptance criteria are documented following the guidelines of ISO-12207 [**ref`ISO`12207**] and MIL-STD-498 [**ref`MIL`STD`498**] (from which ISO-12207 originated). The document follows the **SSS** DID [**ref`SSS`DID**], with a few minor tailoring changes.

The first format tailoring change, if allowed by the customer or sponsor, allows for the system interfaces to be specified before the system capabilities. This follows standard structured design practice (e.g. Yourdon's Structured Method) whereby the system context is provided before the design itself. The net result of this change is that system capabilities are presented in section 3.3 instead of 3.2, as prescribed in the **SSS** template, and external system interfaces are described in section 3.2 instead of section 3.3 as prescribed in the **SSS** template. This allows the data inputs to the system to be defined *before* they are used in the capabilities section. If this artifact does not swap these sections, it is recommended to the reader that they read the external interfaces section before the capabilities section in order to have a better understanding of the system context and external interfaces before delving



into the capabilities, which act upon the data received from and provided to the external interfaces.

The second format tailoring change relates to placement of general material within the document. The qualification provisions and traceability details, if applicable, are listed with each requirement. This formatting, which is allowed in the **SSS DID** [ref***SSS*****DID**], allows the reader to view all relevant information for each requirement in a single location, rather than requiring constant page turning. This information may be duplicated in Sections 4 and 5, respectively, but if done this way, it can be generated automatically to prevent manual duplication errors. The table of acronyms is also listed in chapter two, versus the notes section, so that it may be parsed by readers before encountering most of the acronyms.

Otherwise, this document follows the listed **SSS** sub-section order.

Section 1 provides an overview of the system and this document.

Section 2 lists general and application-specific reference documents as well as glossary terms and acronyms.

Section 3 details the specifications for the system.

Section 4 maps the specifications to quality provisions.

Section 5 traces specifications to the original source.

Appendices if needed, provide additional information as may be needed.

1.3.3 Document Version Information

This document was produced in $\text{\LaTeX}$ and *BibLaTeX/Biber*. The editing and document preparation were performed using $\text{\LaTeX}$ version 2.9 with the build option [$\text{\LaTeX} \Rightarrow \text{PS} \Rightarrow \text{PDF}$]. The $\text{\LaTeX}svn-multi$ package was used to glean SVN tracking information, when files are stored in an “SVN” version control system. The style **KNEADdocument** was used to provide the $\text{\LaTeX}$ and *BibLaTeX/Biber* formatting details.

This revision of this document has the following properties:

1.3.4 Specification Formatting

The specifications are listed and numbered by document sections. The fully qualified specification numbers include the sub-section in which it is contained. These specification numbers are tied to the document level thus they are numbered from 1 to N for each sub-section of the requirements section. This is done to allow for additions within a sub-sections without affecting the numbering in other sub-sections.



Tracking Item	Data
Repository	
Author	KneadProject
Revision	142
Rev Date	2025-08-03 20:42:20Z
Print Date	06 Dec, 2025 15:29
KNEADdocument Version	1.02
KNEADdocument Date	2024/12/21

Once a specification has been added, it cannot be deleted, only its status may be changed to “inactive” or “deleted” to preserve numbering. The table format also allows for grouping of related specifications. These grouped specification are also numbered sequentially and are also not removed if made inactive. Rather, they are marked in a “strike-though” font to denote that they are inactive.

This document allows for marking changes to specifications. All specifications may be marked with a change bar. This generally implies that one or more parts of a specification changed from the prior revision. A note should be provided to indicate the reason for the change, and when, so that future versions of the document, which do not include the change bar, still have rationale included for the current value.

The system specifications are listed in a common table format as shown in Requirement 1.3.1.



Specification 1.3.1 Specification Table Format

Text	The system requirements are listed in a common table format with the following parts:
Specifics	1. The first row of a table provides a unique number and a title for the requirement. 2. The second row of the table provides the text of the requirement. This may extend to the next row for multi-part requirements. See the notes for more information on this practice. This line delineates KPP or KSA requirements, by adding “KPP” or “KSA” to the left side of this row. 3. The next row of the table provides a list of the requirement specifics for multi-part requirements, or is omitted for single-part requirements. 4. The next row of the table provides the status for the items in the table. 5. The next row of the table provides the acceptance criteria. This row follows the form of “This requirement shall be verified by $V \in \{\text{inspection, demonstration, test, analysis}\}$”. 6. The next row of the table provides the traceability of the requirement, which is a higher level document that calls out the need for a requirement. The structure of traceability is expected to be of the form “This requirement is derived from MIL-STD-498 [ref“MIL”STD”498] and ISO-12207 [ref“ISO”12207]”. Note that the source is expected to be listed in the reference documents section. 7. The final row of the table provides, if applicable, notes for the requirement that are not a formal part of the requirement but provide supporting information regarding the feature.
Status	All This format is active for all requirements in this document.
Acceptance	This specification is not a testable requirement for the system; it is for demonstration purposes only.
Traceability	N/A There is no traceability for this requirement.
Notes	1. This table is generated using a L^AT_EX command. 2. The multi-part form is used here so that all of the rows are displayed. 3. This formatting is not a testable requirement on the system, but rather, shows how the requirements are depicted in the document. 4. The multi-part format allows for common items to be grouped together for conciseness and clarity. Each line of the multiple-parts is to be considered as a separate requirement, with a unique identifier (ID) that includes the point number added to the base requirements number. For example, requirement X.Y.Z with 3 parts would be unique IDs X.Y.Z.1, and so forth. 5. The multi-part format is useful for tightly-coupled items. For example, a display monitor has specifications for diagonal size, resolution, brightness, contrast ratio, etc. By placing all such requirements into a single spot helps ensure that all of the closely related requirements are considered together.



The status designations for each specification $\text{Status} \in \{(\text{T})\text{hreshold}, (\text{O})\text{bjective}, (\text{I})\text{nactive}, (\text{D})\text{eleted}\}$ are based on the following criteria.

- (T) - **Threshold** Items marked “(T) - Threshold” are driven by the project threshold needs that must be met in the specified phase.
- (O) - **Objective** Items marked “(O) - Objective” are objective goals of the system in the specified phase. These requirements may stay (O) for all listed phases or may transition from (O) to (T) in future phases. This provides hints as to future expansion of system capabilities so the design can account for the feature without significant later rework.
- (I) - **Inactive** Items marked “(I) - Inactive” are requirements that are not currently to be met by the system in the specified phase. Unlike “(O) - Objective” requirements, (I) requirements may be in limbo in terms of certain details but their inclusion may also provide hints as to future expansion of system capabilities.
- (D) - **Deleted** Requirements that are not to be met by the system are marked by “(D) - Deleted”. Use of this status, vice removal of the requirement text, preserves the numbering of subsequent requirements and notes that the requirement once was invoked. The rationale for the deletion should be included in the notes section.

The status may include the applicable versions in which the required feature is supported. For example, when a phased development approach is used, a format of [Phase-X] Status can be employed, with multiple phases listed, to indicate when the requirement is to be included in the system.

External tools have been written that allow for automatic generation of other documentation. Specially, data for chapters and appendices that follow the requirement specifications, can be gleaned automatically to ensure integrity between the sections of the documents. In addition, the listing of **KPP** and **KSA** values into a “B-spec” can be automated. Finally, the full set of requirements, and the associated attributes, are exported to a comprehensive .CSV file for import into **MBSE** tools such as **DOORS™** or **CAMEO™**.

This table approach offers other advantages besides automated parsing for import into tools. As can be seen in Table 1.3.1, and in all the specifications, this format groups all information for each specification into a separable and easily viewed structure. The document sections and subsections provide a logical grouping of the specifications but the table allows all pertinent information to be grouped, vice being split across major sections of the document. This grouping allows for easier presentation since each grouping is similar to a “PowerPoint” presentation slide. And, as will be seen in Section 1.3.5, it can help the writer organize specifications. The approach also allows for a “List of Specifications” to be generated. Each table is listed in the list of specifications so that each high level grouping can be quickly located from the list. Of course, the tables are located in the appropriate sections as noted in Section 1.3.2 so they can be found in that manner as well.



Another major advantage of the table format is the “Notes” section. As specifications are developed, there will be many issues to be resolved. And, once issues are clarified, tracking the rationale for the decision is just as important as recording the answer [ref‘**Brooks**‘**MMM**]. Thus the notes section helps the reader and the writer. The writer has a logically grouped place to put notes for each specification and the reader can easily find them without having to refer to footnotes, separated sections, or external documentation.

1.3.5 How To Read Specifications

System Performance Specification (**SPS**) documents, by their very nature, are a collection of independent but interconnected facts. Systems require interfaces from which inputs are consumed and to which outputs are produced. These input data are transformed by the capabilities to produce the outputs. The entire system has myriad other requirements ranging from data formatting through physical limits on things like the enclosure and packaging. Finally, at the system level, specifications need to dictate *what, and, how well* a system must perform. Likewise, in order to separate documentation functionality, the **SPS** should not state explicitly *how* the system is to be formed, except in very special circumstances. Given the disparate nature of the requirements, system performance specifications can be hard to digest.

System / Sub-system Specification (**SSS**) and Software Requirements Specification (**SRS**) documents suffer from many of the same issues as do **SPS** documents. An **SSS** turns the performance specification into a first level design. Where the **SPS** has disparate performance requirements, the **SSS** has disjoint hardware and software configuration items listed as well as a mapping of the two items on to, and in to, each other. For the requirement management function, each element of the system design in the **SSS** traces back to the overall **SPS** specifications. Only at the **SRS** level do the requirements start to focus on a single item. Thus, the **SSS** and **SRS** level documents describe share a common contextual issue with the **SPS** document.

An understanding of the documents’ structures is needed to help parse the information. The developers of MIL-STD-498 [ref‘**MIL**‘**STD**‘**498**], however, understood this and devised a format that can help manage the information overload. The method by which this is accomplished is to organize requirements into eighteen specific groupings. By understanding these groupings, a reader can improve their understanding of the system described by the requirements by understanding that specific information is listed in specified sections.



These documents follow a *read-forward* mentality so that base information is provided before it is actually needed.¹ For example, the referenced documents (and in this document the list of acronyms and glossary terms) are provided in the second chapter, before most items are actually cited. By presenting this information to the reader before citation, the format allows the reader to glean upfront information about the kind of things that will be covered later on in the document. Presentation of the referenced material ahead of the document body also allows the reader to have a priori information before encountering the symbol in the text. By understanding these formatting clues, the reader is able to be prepared for what is coming further down the road in the document.

Another reason for this organization is that different readers process information differently. There is no one format that will be best for all readers. By having the information in standardized sections across all such performance specifications, however, readers can use the document as a reference as needed. As an example, Section 3.1 of these documents will always define the system states and modes. If a reader wants to look up this information, it is always in that section. And, since it is listed first, all ensuing sections can reference states and modes in order to qualify their requirements. Likewise, having external interfaces presented *before* the capabilities means that the capabilities can define the transformations without having to worry about how the data is ingested into the system; the data is already “in the system” from a reader’s point of view when the data is needed to define the transformation.

This separation between, and the presentation order of, the data and processing is important when the same data supports multiple capabilities. A hierarchical description of “derived” requirements often would have a capability definition leading to the requirements for the data needed by the capability. In the case, which occurs often, where the data is used in two different transformation capabilities, a hierarchical approach of capability leading to external interfaces is met with the problem of which capability gets to have the data interface in its tree. This approach also leads to the dilemma of what to do when that first capability is no longer needed by the system but the second capability, and thus the data defined under the first capability tree, still *is* needed by the system. Just as loosely coupled software is more maintainable, so are loosely coupled requirements.

A final note about reading actually comes from ideas about how to write a requirement

¹In fact, the read-forward philosophy extends across the documents as well but that topic is beyond the scope of this discussion.



document. The document format dictates a linear flow from start to finish. A reader often reads a document the first time from front to back; this is why the *read-forward* approach works.

The writer, however, is *not* constrained in the same linear manner. Thus, while writing requirements about a capability, the idea for a new state or mode may arise. The writer can easily jump to the states and modes section in order to add in the information and then return to complete the capability that led to the new state or mode, safe in his knowledge that when the reader sees this new specification, they should have already seen the newly defined state or mode that is used to qualify the requirement since the state or mode was already presented in an earlier section.

Readers can use this information to enjoy the fact that the writer jumped around in the writing phase to save the reader the same effort when trying to read the document. By understanding the sections and the expected contents of the sections, which are defined in the **SSS DID** [ref*SSS*DID] and the related MIL-STD-498 [ref*MIL*STD*498] document templates, a reader can read cover-to-cover, or jump to the needed information quickly, knowing that the writer put the information in the specified sections to make finding the information easier for the reader.

1.3.6 Specification Traceability

A project typically has several levels of statements of what needs to be done. Often an Operational Concept Description (**OCD**) is developed to provide a high level description of the system to be developed and its expected uses. A set of documents that follows the MIL-STD-498 [ref*MIL*STD*498] **DID** formats is often developed for a project. A Statement of Work (**SOW**) or a Statement of Objectives (**SOO**) is often developed to direct contractors on a project. While a **SOW** or **SOO** is supposed to be more contractual statements of tasks and objectives rather than actually trying to specify what needs to be built, these documents often, however, do include system requirements. A better method is to have the **SOW** or **SOO** reference a formal **SPS** so that the full scope of the system can be defined. Situations for every project differ so the main thing to understand is that there will be many documents and that their contents need to be related to each other.

Given the number of specifications for a system, and all the levels at which the specifications may be written, understanding if all needs of a system are being met is critical. Different documents have different levels of specifications and design materials. A mapping



between the specifications at all of the different levels is essential to make sure that the design meets all of the stated needs of the system and that the system does not include capabilities that are not required.

As a system performance specification, the **SPS** is the highest level of specification of systems requirements. This document defines *what* the system needs to do without saying *how*. Thus, in general, there should be nothing higher to which the requirements in a **SPS** can be traced. In practice, however, some document such as an **OCD** for the system, informal customer requirements documents, or a **SOW** or **SOO** for the project may be provided that indicates some of the system needs. If the specifications in the **SPS** meet all the use cases in the **OCD** then the system meets the needs but only if the use cases are all inclusive. Likewise, if the **SOW** or **SOO** tries to list things the system needs to do, these needs must be tracked. And, of course, statements of need in any straw-man requirements documents need to be met. Once the fully developed **SPS** specifications are captured then the higher level document(s) be examined to make sure that, at a minimum, all of those things listed are captured in the **SPS**. This process ensures that the **SPS** is compliant with the other “defining” stakeholder documents.

To ensure coverage compliance, each of the higher level needs, whether implied or explicitly stated, needs to be mapped to the **SPS** specification(s) that cover each given need. This is expected to result in a one-to-many relationship where many of the **SPS** specifications are mapped to a single upper level need. And, each specification in the **SPS** can be mapped to multiple needs depending on the independence of the needs. The details of how to do this mapping, which is best handled through some relational database tool (e.g., **MBSE** tools such as **DOORS™** or **CAMEO™**), are beyond the scope of this introduction. The important thing to note is that this traceability determines if the **SPS** covers the known needs of the system. Since a well-formed **SPS** includes many more facets than an **OCD** or **SOW/ SOO**, there may be orphan **SPS** requirements that do not map directly to the higher level documents. However, there can be no orphan needs from the stakeholder documents that do not map to the **SPS**. The key here is to perform a mapping between the **SPS** and any higher level stakeholder documents to ensure that the specifications of the **SPS** provide compliance to the higher document(s).

Another way of saying this is to summarize the overall design philosophy as follows: a level of design should be carried out and mapped to higher level artifacts rather than simply “deriving” requirements from the higher level documents. This mapping process, thus, is



not a way to derive a fully defining set of requirements for the **SPS**. The act of “deriving” specifications of system from a non-specification document such as a **OCD**, **SOW**, or **SOO** does *not* ensure that all the true system needs are captured. In fact if this approach is followed then often many requirements are missed because of the incomplete nature of the **OCD**, **SOW**, or **SOO** list of system needs.

An **SPS**, just like any level document, needs to be developed using domain knowledge and by following best practices and a well-defined documentation format. All of the things that must be considered for defining a successful system need to be included in the **SPS**. Note that, obviously, if the **SOW** did all of this then the **SOW** would be the **SPS**. But, in practice, this rarely ever happens, nor should it. The **SOW** or **SOO** are programmatic level documents that list things to do to build the system; they are not supposed to define the system. This is the job of the **SPS**.

The system / sub-system segmentation specification (**SSS**) is the second level of specification of systems requirements. The **SSS** starts to define *how* the system will meet the needs and should include a top-level system decomposition (or segmentation) of capabilities from the **SPS** to the sub-systems of the system. In a typical documentation set, there should be a higher-level document (i.e. the **SPS**) to which the specifications in the **SSS** can be traced. The goal here is to ensure compliance with the higher level document much as was done with the **SPS** and higher level programmatic documents. If the design specifications in the **SSS** cover all the performance specifications in the **SPS** then the system design covers the documented system needs. This does not mean that the system *will* meet the performance specifications, it just means that there are no obvious holes. And, of course, if some **SPS** specifications are not covered in the **SSS** then those specifications obviously cannot be met by the system design.

Since the **SPS** and **SSS** documents are all explicit statements of requirements, there can be no gaps between the specifications in them. If there is no mapping from at least one **SSS** requirement to each of the **SPS** requirements then the system cannot meet the stated requirements of the **SPS**. While the goal here is to ensure that the specifications of the **SSS** provide coverage to the higher document, true compliance can only be determined through design analysis and testing efforts. And, if there is an orphan **SSS** requirement then the questions must be asked: “Why is this requirement included?” and “Can this requirement be deleted?”. Taken together, the **SPS** and **SSS** documents can form the basis for a well-formed design artifact set: References to higher-level artifacts document where the system



goals came from and the code (and hardware) level artifacts (source code and CAD models) should describe the final product details. Coupled with appropriate test documents, this level of design process and documentation generation should adequately allow for successful development while preserving the architectural aspects for future revisions and modifications.

Because this is the overall system performance specification, this document may provide traceability to miscellaneous project documents. This allows for tracking of related doctrine, vendor, and draft specification requirements as the document is being created.



CHAPTER 2

References

This section provides a list of referenced items for this document.

2.1 Acronyms and Abbreviations

This section defines acronyms and abbreviations used in this and related documents.

Table 1: Acronym Definitions

Acronym	Definition
.AUX	LaTeX Auxiliary Files
.CSV	Excel Comma Separated Variable
.PDF	Portable Document Format
.PS	Portable Document Format
ABIS	Automated Biometric Identification System
BioAPI	BioAPI
DIDs	Data Item Descriptions
DOORS TM	DOORS
Gantt	Gantt Chart
Garmin	Garmin Inc.
HMI	Human-Machine Interface
JIDA	Joint Improvised-threat Detection Agency
KPP	Key Performance Parameter
KSA	Key System Attribute
MISO	Military Information Support Operations
OCDs	Operational Concept Descriptions
SOO	Statement Of Objectives
SOW	Statement Of Work
SPSs	System Performance Specifications
SRD	System Requirements Document
SRSs	System Requirements Specifications
SSSs	System / Subsystem Specifications
Acronym definitions continue on next page	



Acronym definitions – continued from previous page	
Acronym	Definition
STS	System Test Specification
TOC-L	Tactical Operations Center - Light
UAV	Uncrewed Aerial Vehicle
USAF	United States Air Force
USArmy	United States Army
USMC	United States Marine Corp
USN	United States Navy
USSF	United States Space Force
...TBD...	...TBD...
End of acronym definition table	

2.2 Glossary and Definitions

This section defines glossary terms used in this and related documents.

Table 2: Glossary Terms and Definitions

Glossary Term	Definition
STM32F429I	Micro-controller board has all component fit onto one board.
Customer	The professor that is view the grading all assignments.
End of glossary terms table	

2.3 Referenced Documents

This section lists the referenced documents for this document. The references are categorized into two categories:

External Documents not directly associated with this project.

Project Documents that are directly associated with this project.

2.3.1 External Documents

2.3.2 Project Specific Documents



CHAPTER 3

Requirements

This section provides the requirements that drive the design and implementation of the Balance System. These specifications are divided into the major segments of the system. Each requirement is listed in the segment that provides the specified capability, thus this section provides an immediate mapping of implementation requirements to the segment in which each requirement is met. Each requirement also includes traceability to high-level requirements that drive the specific capability. Validation methodology is provided here but verification traceability is provided in the **STS** artifacts.

The requirements also are specified in an order that generally allows for all precursor requirements to be stated before they are needed by a successor requirement. Thus, States and Modes are defined at the onset so that they can be used to regulate when external interfaces and processing steps may occur. Likewise, external interfaces are described so that the data from the interfaces may be used in, or created by, the ensuing processing. Once the processing is specified, the internal interface and data requirements are listed, showing how the overall system segments tie together. The remainder of the sections follow a somewhat similar pattern, but these latter sections contain disparate requirements that are separated and organized in a standard way so the contents can be easily scanned to locate specific requirements based on their type and expected location within the **SSS**.

These specifications also include qualifications for both Threshold (must meet) and Objective (want to meet) requirements for Balance System. The reader is cautioned to ensure that the requirement details be understood for the two modifiers.

3.1 States and Modes

This section lists the states, sub-states, modes, and sub-modes that are provided by the system. While these terms can be construed in many ways, for this document, the following meanings are used:

States are the basic configurations of the system.

Sub-states are the effective state of being for the system.



3.1.1 States

A summary of the states is provided in Table 3. See the formal specifications, if applicable, in the following sections for formal statement of the state requirements, and accompanying notes that provide further clarification on the meanings of the states.

STATES	
State Name	Summary
State 1	Main Page 3111
State 2	Help Page 3112
State 3	Main Game Mode 3121
State 4	Balls movement 3131
State 5	Ball matches target 3132
State 6	Displaying level 3133
State 7	Displaying time 3134
State 8	Changing levels 3141
State 9	Failed level 3142
State 10	Hard turns 3143

Table 3: Summary of States for Balance System

Specification 3.1.1.1 State One	
(KPP) Text	The Balance System will have an opening screen that the game should also start with. This will allow the user to select viewing details of how to play the game or just to play the game.
Status	Phase 1 Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	SPS This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.



Specification 3.1.1.2 State Two

(KSA) Text	The Balance System shall have a separate page that give clear indication on how the game shall be played. This page shall take the whole LCD screen to display.
Status	Phase 1 Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	SPS This requirement is referenced in the SPS sheet.
Notes	N/A This requirement is referenced in the SPS sheet.

Specification 3.1.1.3 State Three

Text	The Balance System shall have a consistent location where the ball within each level will have the ball placed. This place will be noticeable and be accessible for any player to move the ball in the correct location..
Status	Phase 1 Levels must transition to the next live after every win.
Acceptance	This requirement shall be verified by demonstration.
Traceability	SPS This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.4 State Four

Text	"The Balance System shall have a transitional model that would change the position on the ball based off the MEMs reading. The reading from the Mem control should describe the movement of the ball"
Status	Phase 1 Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	SPS This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.



Specification 3.1.1.5 State Five

Text	The Balance System shall a win page once the user matched the write location for where the ball should be landing.
Status	Phase 1 Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	SPS This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.6 State Six

Text	The Balance System shall have a level score while the game is running. This would be a running counter that is displayed once the user starts the game.
Status	Phase 1 Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	SPS This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.7 State Seven

Text	The Balance System shall provide a timer on the top left of the game to keep track of level to keep track of time per level. This is an ideal location for the user see there time limit.
Status	Phase 1 Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	SPS This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.



Specification 3.1.1.8 State Eight

Text	The Balance System shall change levels once the user reached a certain level.
Status	Phase 1 Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	SPS This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.9 State Three

Text	The Balance System shall go back main menu once the user does not make the mark of the user.
Status	Phase 1 Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	SPS This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.10 State Ten

Text	The Balance System shall discard sharp changes while the microcontroller is being tilted.
Status	Phase 1 Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	SPS This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.



3.2 External Interfaces

The external interfaces for this system are shown in Figure 2. The requirements for these interfaces are described in more detail in the following sections.

Network The network(s) that connect to the Balance System, § 3.2.1

Power The network(s) that connect to the Balance System, § 3.2.1

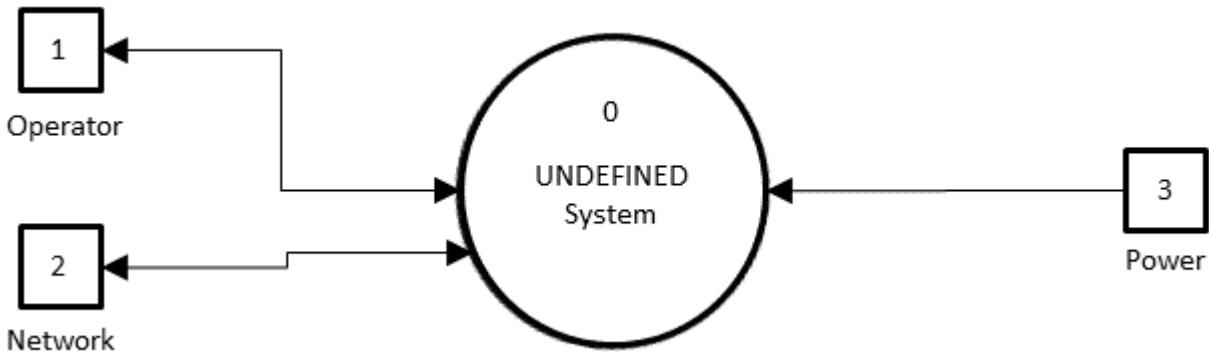


Figure 2: System Context Diagram (DFD-C)

3.2.1 Network Interfaces

Specification 3.2.1.1 Approved Network	
(KPP) Text	The current network would not be needed for completion of the system.
Status	Phase 1 The system must work independent from the main system.
Acceptance	The system would not need to be dependency from any network communication.
Traceability	N/A The base requirement comes from the STP document.
Notes	1. N/A



3.2.2 Power Interfaces

Specification 3.2.2.1 Power	
(KPP) Text	All Balance System variants shall be capable of connecting to ...TBD... Power.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This requirement is a base requirement.
Notes	1. N/A

3.3 Capabilities

This section defines the capability areas for the Balance System. The segment design is structured to meet the requirements as specified in the ...TBD... artifacts. Each area provides a subset of the overall capabilities for the Balance System segments. These segments are shown in Figure 3, are summarized below, and are more fully specified in the following subsections.

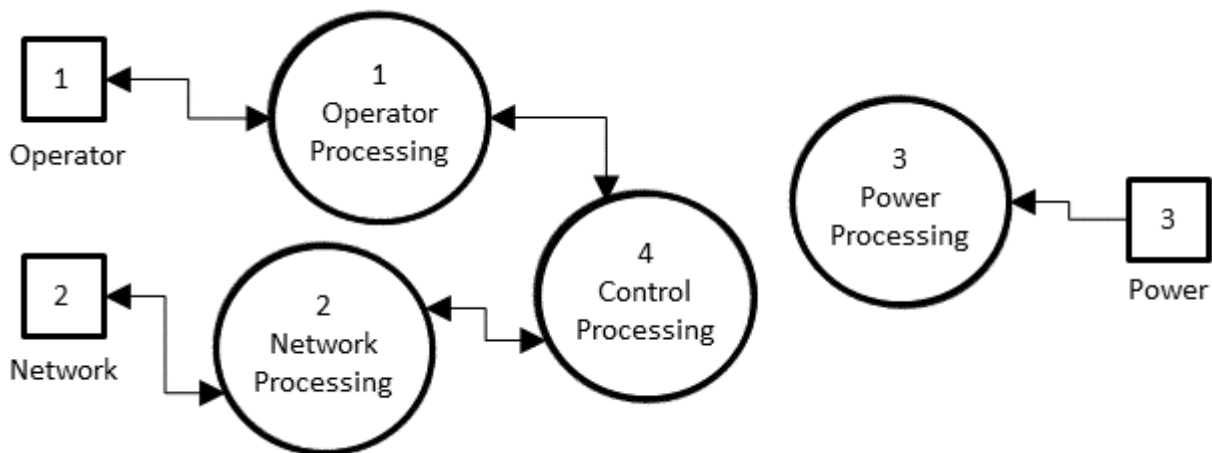


Figure 3: System Top-Level Diagram (DFD-0)

The capability requirements for these segments are described in more detail in the following sections:

Operator Processing handles the **HMI** interface to the operator and provides overall control and configuration to the Balance System, § 3.3.1.

Network Processing handles the network interface, § 3.3.2.



Power Processing handles the power input and conversions as necessary, § 3.3.3.

Control Processing handles all major capability control, § 3.3.4.



3.3.1 Operator Processing

The operator requirements for Balance System are listed below.

Specification 3.3.1.1 Power	
(KPP) Text	All Balance System variants shall be capable of ...TBD... operator inputs.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files, e.g., OperatorInputs.tex and OperatorOutputs.tex, as needed. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.

Specification 3.3.1.2 Power	
(KPP) Text	All Balance System variants shall be capable of ...TBD... operator outputs.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files, e.g., OperatorInputs.tex and OperatorOutputs.tex, as needed. Just use the RequirementNumberAM command to keep numbers correct if subsubsections are added.



3.3.2 Network Processing

The network requirements for Balance System are listed below.

Specification 3.3.2.1 Network Types	
(KPP) Text	All Balance System variants shall be capable of ...TBD... network types.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders, e.g., NetworkTypes.tex, NetworkInputs.tex, and NetworkOutputs.tex, etc. as needed. Just use the RequirementNumberAM and RqtNumber-Base commands to keep numbers correct if subsubsections are added.

Specification 3.3.2.2 Network Inputs	
(KPP) Text	All Balance System variants shall be capable of ...TBD... network inputs.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders, e.g., NetworkTypes.tex, NetworkInputs.tex, and NetworkOutputs.tex, etc. as needed. Just use the RequirementNumberAM and RqtNumber-Base commands to keep numbers correct if subsubsections are added.

Specification 3.3.2.3 Network Outputs	
(KPP) Text	All Balance System variants shall be capable of ...TBD... network outputs.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders, e.g., NetworkTypes.tex, NetworkInputs.tex, and NetworkOutputs.tex, etc. as needed. Just use the RequirementNumberAM and RqtNumber-Base commands to keep numbers correct if subsubsections are added.



3.3.3 Power Processing

The power requirements for Balance System are listed below.

Specification 3.3.3.1 Power Voltage	
(KPP) Text	All Balance System variants shall be capable of ...TBD... power voltage(s).
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders as needed. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.

Specification 3.3.3.2 Power Current	
(KPP) Text	All Balance System variants shall be capable of ...TBD... power current.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders as needed. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.



3.3.4 Control Processing

The control requirements for Balance System are listed below.

Specification 3.3.4.1 Control One	
(KPP) Text	All Balance System variants shall be capable of ...TBD... control one.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders as needed for areas of control capabilities. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.

Specification 3.3.4.2 Control Two	
(KPP) Text	All Balance System variants shall be capable of ...TBD... control two.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders as needed for areas of control capabilities. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.

3.4 Internal Interface Requirements

This section provides the internal interface requirements. These requirements for these interfaces are described in more detail in the following sections:

Internal Interface Requirement One stuff

Internal Interface Requirement One more stuff

3.5 Internal Data Requirements

This section provides the internal data requirements. The Balance System capability is segmented into the following specification groups:

Data Storage provides the data storage requirements, § 3.5.1.

Report Logs provides the report log requirement, § 3.5.2.



3.5.1 Data Storage

Specification 3.5.1.1 Data Storage	
(KSA) Text	All Balance System variants shall store digital files received for transmission using 2 TB of internal storage.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This is a base requirement.
Notes	1. N/A

Specification 3.5.1.2 Information Transport	
(KSA) Text	All Balance System variants shall be able to manually upload and download digital files using external SD card, CD, and DVD formats.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This is a base requirement.
Notes	1. N/A

3.5.2 Report Logs

Specification 3.5.2.1 Report Logs	
(KSA) Text	The system shall locally store reports for up to 12 months and be capable of exporting in .txt, .csv, and .xml formats.
Status	Phase 1 T=O
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This is a base requirement.
Notes	1. Where does this information come from? ...TBD....

3.6 Adaptation Requirements

Later on the next idea is add legs to the system.

3.7 Safety Requirements

This section lists the safety requirements for the system. The Balance System capability is segmented into the following specification groups:



Electromagnetic Radiation describes the safety requirements pertaining to the presence of EMR, § ??.

3.8 Security and Privacy Requirements

This section provides the security and privacy requirements for Balance System. The Balance System capability is segmented into the following specification groups:

Security Requirements provides the physical and cyber security requirements of the system, § 3.8.1.

Privacy Requirements provides the privacy requirements of the system, § 3.8.2.

3.8.1 Security Requirements

3.8.1.1 Physical Security

Specification 3.8.1.1.1 Anti-Tamper	
Text	The Balance System shall deter all unauthorized alterations, countermeasure development, and system exploitation.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by testing.
Traceability	N/A This is a base requirement.
Notes	1. N/A

3.8.1.2 Cyber Security

This section is provided for future expansion.

3.8.2 Privacy Requirements

This section is provided for future expansion.

Specification 3.8.1 Privacy	
Text	The system shall ... TBD
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This is a base requirement.
Notes	1. ... TBD



3.9 Environmental Requirements

This section defines the environmental requirements for Balance System.

3.10 Technology Resource Requirements

This section provides the overall technology resource requirements for the system. These capabilities are divided into the following sections:

Hardware details about the hardware to be used.

Software details about the software to be used.

Communications details about the communications to be used.

Other details about other technology resource requirements not covered above.

Utilization details about the resource utilization.

3.11 System Quality Requirements

This section specifies the Balance System quality requirements.

3.11.1 Quality Systems

Specification 3.11.1.1 Development Quality	
Text	The system design and development shall follow the implementers' certified processes.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by inspection.
Traceability	N/A This requirement is a base requirement.

3.11.2 Operational Quality

Ensure that the system would load properly connects with the correct subsystem.

3.11.3 Quantitative Metrics

3.11.3.1 Object Detection Metrics

3.11.3.2 Object Identification Metrics

3.11.4 Qualitative Metrics

3.11.4.1 Media Selection Metrics

This Section is ...**TBD**....



3.12 Design and Construction Requirements

This section provides the Balance System design and construction requirements.

3.12.1 Regulatory Restrictions

This section is included for future expansion.

Specification 3.12.1.2 Proprietary Components	
Text	All Balance System variants shall include only software components that are open source or to which the developer has unlimited rights.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by inspection.
Traceability	N/A This requirement is a base requirement.
Notes	1. The inspection is of the software design documents and build scripts to ensure that all code meets this requirement.

3.12.2 Design Defences

This section provides the construction constraints.

This section is provided for future expansion.

3.12.3 Construction Constraints

This section provides the construction constraints.

This section is provided for future expansion.

3.13 Personnel Requirements

This section is provided for future expansion.

3.14 Training Requirements

3.14.1 Manuals

Specification 3.14.1.1 Operator's Guide	
Text	The Balance System training shall provide an operator's manual.
Status	Phase 1 T=O
Acceptance	This requirement shall be verified by inspection.
Traceability	N/A This is a base requirement.
Notes	1. N/A



3.14.2 Materials

Specification 3.14.2.1 Training Materials	
Text	The Balance System training shall provide all training course materials.
Status	Phase 1 T=O
Acceptance	This requirement shall be verified by inspection.
Traceability	N/A This is a base requirement.
Notes	1. N/A

3.14.3 Courses

Specification 3.14.3.1 Training Courses	
Text	The vendor will provide specific training and course materials and programs of instruction.
Status	Phase 1 T=O
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This is a base requirement.
Notes	1. N/A

3.15 Logistics Requirements

3.15.1 Transportability

Specification 3.15.1.1 Transportability	
(KSA) Text	All Balance System variants shall be transportable by air, ground, and maritime resources.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This is a base requirement.
Notes	1. N/A

3.16 Packaging Requirements

3.16.1 Shipping Constraints

This section is provided for future expansion.



3.17 Other Requirements

3.17.1 Broadcast Playlist Manager

Specification 3.17.1.1 Playlist Manager	
(KSA) Text	All Balance System variants shall interface with an approved external broadcast playlist manager that will provide the operator the ability to manage programming in real-time.
Status	Phase 1 T=O
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This is a base requirement.
Notes	1. N/A

3.17.2 Information Exchange

Specification 3.17.2.1 IP Data	
(KPP) Text	All Balance System variants shall be capable of disseminating data on an approved network with a reasonable response time of less than four (4) hours.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	N/A This is a base requirement.
Notes	1. N/A

3.18 Precedence of Requirements

3.18.1 Safety

Specification 3.18.1.1 Safety Requirements Precedence	
Text	All Balance System variants shall meet safety requirements listed in Section 3.7 before all other requirements.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by inspection.
Traceability	N/A This requirement is a base requirement.
Notes	1. Obviously safety is of utmost importance. 2. The inspection is of design notes and rationale whereby design decisions relating to precedence are recorded.



3.18.2 Security and Privacy

Specification 3.18.2.1 Security Requirements Precedence	
Text	All Balance System variants shall meet security requirements listed in Section 3.8.1 before all other requirements with the exception of safety.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by inspection.
Traceability	N/A This requirement is a base requirement.
Notes	1. Security trumps privacy since good security should help ensure privacy. 2. The inspection is of design notes and rationale whereby design decisions relating to precedence are recorded.

Specification 3.18.2.2 Privacy Requirements Precedence	
Text	All Balance System variants shall meet privacy requirements listed in Section 3.8.2 before all other requirements with the exception of safety and security.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by inspection.
Traceability	N/A This requirement is a base requirement.
Notes	1. Privacy is trumped by security since good security should help ensure privacy. 2. The inspection is of design notes and rationale whereby design decisions relating to precedence are recorded.

3.18.3 Other

Specification 3.18.3.1 Other Requirements Precedence	
Text	All Balance System variants shall meet with equal precedence all other requirements not pertaining to safety, security, and privacy.
Status	Phase 1 Threshold
Acceptance	This requirement shall be verified by inspection.
Traceability	N/A This requirement is a base requirement.
Notes	1. The inspection is of design notes and rationale whereby design decisions relating to precedence are recorded.



CHAPTER 4

Qualification Provisions

The qualification provisions are listed in the acceptance row of the specifications in Section 3.



CHAPTER 5

Traceability

This section provides a list of the sources, if applicable, for each requirement. This traceability connects the specifications in this document to those presented in higher level sources such as a Joint Urgent Operational Need (**JUON**) document, Joint Emergent Operational Need (**JEON**), or a STATEMENT OF WORK (**SOW**)

The traceability of all specifications from each requirement to its source, if applicable, is listed in the specifications presented in section 3. Traceability from each document to requirements is provided below.

Table 4: Source to Requirement Traceability.

Source Requirement	Traced Requirement
...TBD...:: 4.1.2	??
N/A :: This is a base requirement.	3.5.1.1
N/A :: This is a base requirement.	3.5.1.2
N/A :: This is a base requirement.	3.5.2.1
N/A :: This is a base requirement.	??
N/A :: This is a base requirement.	??
N/A :: This is a base requirement.	3.8.1.1.1
N/A :: This is a base requirement.	3.8.1
N/A :: This is a base requirement.	3.14.1.1
N/A :: This is a base requirement.	3.14.3.1
N/A :: This is a base requirement.	3.15.1.1
N/A :: This is a base requirement.	3.17.1.1
N/A :: This is a base requirement.	3.17.2.1
N/A :: This is a base requirement.	3.14.2.1
N/A :: This requirement is a base requirement.	??
N/A :: This requirement is a base requirement.	??
Continued on next page	



Table 4 – continued from previous page

Source Requirement	Traced Requirement
N/A :: This requirement is a base requirement.	??
N/A :: This requirement is a base requirement.	??
N/A :: This requirement is a base requirement.	??
N/A :: This requirement is a base requirement.	??
N/A :: This requirement is a base requirement.	??
N/A :: This requirement is a base requirement.	3.2.1.1
N/A :: This requirement is a base requirement.	3.2.2.1
N/A :: This requirement is a base requirement.	3.3.1.1
N/A :: This requirement is a base requirement.	3.3.1.2
N/A :: This requirement is a base requirement.	3.3.2.1
N/A :: This requirement is a base requirement.	3.3.2.2
N/A :: This requirement is a base requirement.	3.3.2.3
N/A :: This requirement is a base requirement.	3.3.3.1
N/A :: This requirement is a base requirement.	3.3.3.2
N/A :: This requirement is a base requirement.	3.3.4.1
N/A :: This requirement is a base requirement.	3.3.4.2
N/A :: This requirement is a base requirement.	3.12.1.2
N/A :: This requirement is a base requirement.	3.18.1.1
N/A :: This requirement is a base requirement.	3.18.2.1
N/A :: This requirement is a base requirement.	3.18.2.2
N/A :: This requirement is a base requirement.	3.18.3.1
KPP-1 :: [ref`KNEAD`Manual]	3.1.1.1
KPP-1 :: [ref`KNEAD`Manual]	??
KSA-1 :: [ref`KNEAD`Manual]	3.1.1.2
KSA-2 :: [ref`KNEAD`Manual]	??
Multi-3 :: [ref`KNEAD`Manual]	??
Single-3 :: [ref`KNEAD`Manual]	3.1.1.9



APPENDIX A

Notes

This section provides notes, as necessary, to document the system segmentation specification.



APPENDIX B

Key Performance Parameters and System Attributes

This Appendix provides the key performance parameters and key system attributes, summarized in a short list for easy review.

B.1 Key Performance Parameters

Table B.1: Key Performance Parameter Specifications

Specification	Key Performance Parameter
3.1.1.1	
??	Thingie one. Thingie two. Thingie three.
??	
3.2.1.1	
3.2.2.1	
3.3.1.1	
3.3.1.2	
3.3.2.1	
3.3.2.2	
3.3.2.3	
3.3.3.1	
3.3.3.2	
3.3.4.1	
3.3.4.2	
3.17.2.1	



B.2 Key System Attributes

Table B.2: Key System Attribute Specifications

Specification	Key System Attribute
3.1.1.2	
??	Thingamabobs one. Thingamabobs two. Thingamabobs three.
3.5.1.1	
3.5.1.2	
3.5.2.1	
3.15.1.1	
3.17.1.1	



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